

Introduction to Computer Programming

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Outline

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Greetings, folks!

Hello students!

Welcome to the *Introduction to Computer Programming* course!

Web Resources and More

You should have access to **LUISS Learn**

<https://learn.luiss.it/course/view.php?id=23440>

(Course website to be set up, I'll add useful info shortly ... I promise)

Course Syllabus

<https://www.luiss.it/cattedreonline/corso/TDS03/0/LAIMBASE/2024>

Rooms and Timing

<https://pianificazionespazi.luiss.it/PortaleStudentiLuiss>

LUISS Learn is our e-Learning platform
<https://learn.luiss.it/>

Important that you get access. You'll find there:

- Slides and other teaching material
- Exercises
- Details on exam dates, reservations, results (besides standard Luiss channels)
- Communications from teacher and teaching assistants

LUISS



Learn



Students Keyless and Luiss accounts Login



Non Luiss accounts Login

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Modalità di accesso

- Gli studenti con un account @studenti.luiss.it devono accedere alla piattaforma cliccando in alto a destra sul link **Students Keyless and Luiss accounts Login** e successivamente su **Riconoscimento facciale per studenti**, utilizzando Keyless. Per configurare Keyless, consulta questa guida.
- In caso di problemi con Keyless potete contattare il supporto all'indirizzo <https://support.keyless.io>.
- Per richiedere il reset della vostra password di un account Luiss, potete scrivere a support@luiss.it.
- In caso di diverse problematiche, potete consultare le [FAQ per studenti](#), le [FAQ per docenti](#), la pagina di tutorial per docenti o, per problemi relativi alle funzionalità della piattaforma, scrivere all'indirizzo elearning@luiss.it.
- Se hai necessità di installare il Respondus Lockdown Browser per eseguire un esame, puoi andare su questa [pagina](#).

To access

- Students with a @studenti.luiss.it account must access the platform by clicking on the **Students Keyless and Luiss accounts Login** link at the top right and then on **Riconoscimento facciale per studenti**. To set up Keyless, see this guide.
- For problems with the keyless authentication, you can log to the site <https://support.keyless.io>.
- To ask a password reset of a Luiss account, you can write to support@luiss.it.
- In case of different problems, you can read the [FAQs for students](#), the [FAQs for teachers](#) or, for problems related to the functions of the platform, write to elearning@luiss.it.
- If you need to install the Respondus Lockdown Browser to take an exam, you can [click on this link](#).

La Luiss mette a disposizione **contenuti didattici in e-learning** volti a supportare la didattica frontale e ad agevolare - in un'ottica di *blended learning* - il lavoro autonomo dello studente necessario ad integrare quanto svolto in classe.

Luiss provides e-learning content to support frontal teaching and facilitating - in a blended learning perspective - the student's independent work to integrate what has been done in the classroom.



Navigation

[Home](#)[Courses](#)

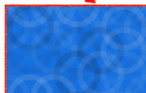
Courses

[Expand all](#)



Dashboard

Recently accessed courses



Management and computer science

INTRODUCTION TO COMPUTER PRO



Connetti alle aule virtuali di lezione - Con

Sale lezioni (da 31 a 40) - Lesson roc



Tutorial per docenti - Tutorials for teachers

Tutorial per docenti sull'uso di Web

After clicking on "My Home"
you'll get a list of the courses
you are enrolled in.

Customise this page



Navigation

Dashboard

Site home

Site pages

My courses

Dipartimento di Impresa e Management



Private files

No files available

Manage private files...



Timeline



No upcoming activities due

Course overview

All (except removed from view)

course name

Card



Personnel

Prof. Dr. Alessio Martino

Assistant Professor of Computer Science

amartino@luiss.it

Office Hours: send me an e-mail so we can arrange a time slot.

Dr. Giorgio Piccardo

Teaching Assistant (TA)

gpiccardo@luiss.it

Office Hours: you can ask info in his first Lab session.

Dr. Fabio Angeletti

Teaching Assistant (TA)

fangeletti@luiss.it

Office Hours: you can ask info in his first Lab session.

Course Info

- The course is intended to teach **basic programming concepts** to students with no prior coding experience, developing an attitude towards **computational thinking**
- It will provide the students with:
 - an understanding of the role that computation can play in solving problems
 - the ability to write small programs that allow them to accomplish useful goals

The course will familiarise the students with computer programming and will cover the following topics:

- Introduction to computation
- Introduction to programming languages
- Python basics
- Variables, data types, expressions
- Control structures
- Repetition structures
- Input / output
- Functions and modules
- Recursion
- Strings
- Lists
- Dictionaries
- Object-oriented programming and classes

- Lectures + Lab Sessions

Laptops at your fingertips, always!

- Monday @ 14:45–16:15

Several rooms in Viale Romania
(often) Lab Session with TA

- Tuesday @ 10:00–11:30

A308 (Viale Romania)
(often) Theory lectures & some exercises

- Wednesday @ 15:45–17:15

A211 (Viale Romania)
(often) Lab Session with TA

- **Coding** and **Multiple-choice Tests** (70% of the overall grade – i.e., 50% coding and 20% multiple-choice)
- **Group project** (30% of the overall grade)

In light of the LUISS Continuous Assessment verification method, we will be having 2 *in itinere* exams:

- First Intermediate Test (tentative date: 23/10/2024)
- Second Intermediate Test (tentative date: 26/11/2024)

that replace the **Coding test** and the **Multiple-choice test**.

Students that will not take (or pass) the intermediate tests during the course are required to take the full test at the end of the course.

Details on the Group project approx at Week 6.

Each intermediate test is a short exam on the topics covered so far.

It will consist of a mixture of coding exercises and theory questions (multiple-choice quiz).

Rules (in brief):

- the sum of the coding parts between the two tests yields 31
- the sum of the theory parts between the two tests yields 31
- a score greater than or equal to 16 in each part is enough for successfully passing the coding + theory tests

Extra bonus points:

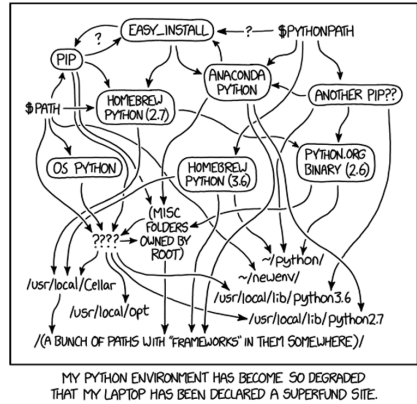
- pass the coding + theory test via intermediate tests, you'll get an extra bonus point
- if you pass the overall exam (coding + theory + group project) within the Winter session, you'll get an another bonus point

Python, Dear Friend

There are just too many ways to use Python in our daily life...



Setting up the Python environment ... ?!



We will use Python 3, no Python 2

As of January 1st, 2020, Python 2 has been officially discontinued

Python 3 can be found at

<https://www.python.org>

along with its documentation at

<https://docs.python.org/3/>

- very rich
- very detailed
- requires some experience

You can install Python via Anaconda



- the most popular **data science platform**
- distribution that includes many many **packages**
- **professional**, with a lot of **functionalities**
- **cross-platform**
- **free** for individuals



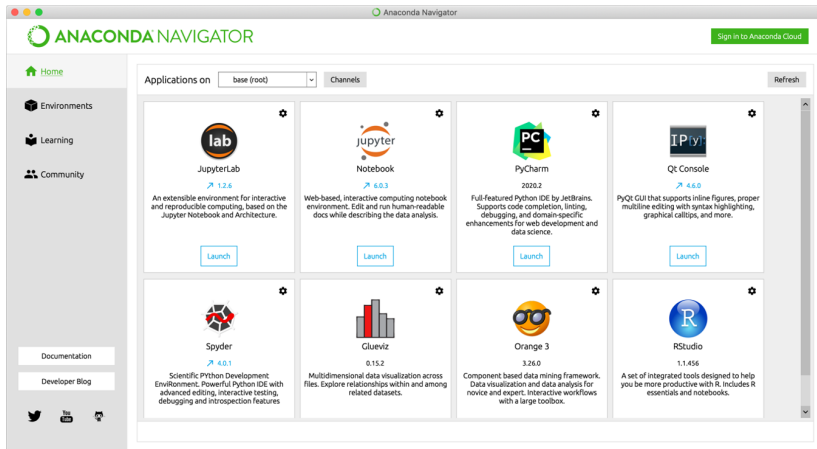
Some Anaconda resources:

- Official website & download:
<https://www.anaconda.com/>
- Documentation:
<https://docs.anaconda.com/anaconda/>
- Installation guide:
<https://docs.anaconda.com/anaconda/install/>
- Community:
<https://anaconda.org>
- Anaconda Navigator:
<https://docs.anaconda.com/anaconda/navigator/>

ToDo:

→ Try to install Python/Anaconda before the next lab session!
You can get help from the TA, should you have any issues.

Like Virgil with Dante, Anaconda Navigator guides you through



A full in-browser alternative: PythonAnywhere

<https://eu.pythonanywhere.com/>

- Work in Python from a browser
- Support both interactive and script mode (files)
- Need to create an account
 - A free account is more than enough for this course
- Good tutorial by Allen Downey
 - <http://www.allendowney.com/wp/books/think-python-2e/>

A shoulder to cry on if your code doesn't work: Python Tutor

<http://pythontutor.com/>

- Observe content of variables
- Observe single execution steps
- Go forward and backwards, even step-by-step

[Get live help](#) in the [Python Discord](#) chat

Python 3.6
([known limitations](#))

```
1 A = [7, 'cat', 3.5, 'dog', 12, 0.3]
2 B = []
3 while len(A)>0:
4     y=A.pop()
5     B.append(y)
```

[Edit this code](#)

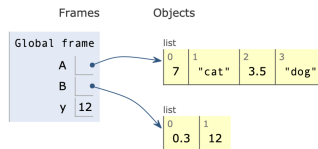
→ line that just executed

→ next line to execute



Step 10 of 21

[Customize visualization](#) (NEW!)



Acknowledgments

- <https://xkcd.com/1987/>
- <https://www.anaconda.com/products/individual>